

E-Commerce Sales Data Analysis

Project Overview

This project analyses E-Commerce sales data to uncover business insights, customer purchasing behaviour, product performance, and sales trends. The analysis helps businesses make data-driven decisions to improve revenue, customer satisfaction, and operational efficiency.

Objectives

- Analyse overall sales performance.
- Identify top-selling products and categories.
- Study customer purchasing patterns.
- Examine monthly and yearly sales trends.
- Discover high-revenue products.
- Visualize key business metrics using charts and graphs.

Dataset

The dataset contains transactional information from an e-commerce platform, including:

- Order ID
- Product Name
- Category
- Quantity Sold
- Sales Amount
- Profit
- Customer Information
- Order Date
- Region/Location

Tools & Technologies

- Python

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Project Workflow

1. Data Collection

Imported the e-commerce sales dataset in CSV format.

2. Data Cleaning

- Checked for missing values.
- Removed duplicate records.
- Corrected data types.
- Handled inconsistent data.

3. Exploratory Data Analysis (EDA)

- Summary statistics.
- Sales distribution analysis.
- Product category analysis.
- Customer behavior analysis.

4. Data Visualization

Created visualizations such as:

- Monthly Sales Trend
- Top Selling Products
- Category-wise Sales
- Revenue Distribution

- Profit Analysis
- Regional Performance

Key Findings

Sales Performance

- Sales showed significant variation across different months.
- Certain periods generated higher revenue than others.

Product Analysis

- A small number of products contributed to a large portion of total sales.
- Some categories consistently outperformed others.

Customer Insights

- Repeat customers generated a substantial share of revenue.
- Customer purchasing behaviour varied by region.

Profitability

- High sales volume did not always result in high profit.
- Some products had strong revenue but low profit margins.

Visualizations Included

1. Monthly Sales Trend
2. Top 10 Products by Sales
3. Category-wise Revenue
4. Profit Distribution
5. Regional Sales Comparison
6. Customer Purchase Analysis

Conclusion

The analysis provides valuable insights into sales trends, customer behaviour, and product performance. These findings can help businesses optimize inventory management, marketing strategies, and customer engagement efforts to improve overall profitability.

Future Improvements

- Sales forecasting using Machine Learning.
- Customer segmentation analysis.
- Recommendation system development.
- Interactive dashboard using Power BI or Tableau.